Regime-Conditional Bias in Fed Funds Futures vs Kalshi Distributions: A Replication Note on Diercks, Katz, and Wright (2026)

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Abstract

Diercks et al. [2026] (henceforth DKW) compare Kalshi-derived FOMC outcome distributions to fed funds futures-implied distributions and report that Kalshi's median and mode have a perfect record relative to the realized federal funds target rate by the day of each meeting since 2022, statistically dominating the futures-implied mean. They argue that fed funds futures distributions systematically (i) over-concentrate on the modal outcome, (ii) exhibit narrower distributional spread, and (iii) under-allocate probability to non-modal tails, due to the binomial-tree assumption that underlies the standard CME FedWatch decomposition. We first replicate their Table 3 Panel B forecast-accuracy result on a like-for-like 27-meeting sample (September 2022 through December 2025) using their own published moments file: Kalshi median and mode MAE replicate to 0.0000 exactly; Kalshi mean MAE replicates within their two-decimal table-display rounding (0.013 vs. their reported 0.010). We then extend their analysis on a 29-meeting sample (Sept 2022 through March 2026, adding the two 2026 meetings DKW's repository now publishes) by operationalizing their Section 5 qualitative distributional claims as three numerical sign-match patterns (mode-gap, spread-gap, tail-gap) and conditioning on a regime classification by minimum pre-meeting Kalshi mode probability. The documented patterns are heavily *regime-conditional*: in the 20 "quiet" meetings the patterns hold at 40/50/50%, indistinguishable from random; in the seven "mixed" meetings they hold at 71/71/71%, just clearing the implicit 70% threshold; in the two "contested" meetings (March 2023 post-SVB, September 2024 50bp surprise) two of three patterns are 2/2. We argue that DKW's qualitative claim is correct for the regime in which it applies but should not be extrapolated to a constant per-meeting bias.

1 Introduction

Diercks et al. [2026] introduce Kalshi macroeconomic prediction markets as a new source of high-frequency expectations data and document several ways in which Kalshi-implied distributions of upcoming FOMC outcomes differ from those derived from federal funds futures. The authors construct daily Kalshi distributions across 11 to 12 absolute-rate strikes per FOMC meeting, applying a volume-weighted last-trade construction with monotonic-CDF cleanup outward from the modal bucket (Diercks et al., 2026, §3). They compare these to fed funds futures-derived distributions constructed under the standard binomial-tree assumption that underlies the CME FedWatch tool, finding [Diercks et al., 2026, Table 3 Panel B] that the Kalshi median and mode forecasts attain mean absolute error of 0.000 on the day of each FOMC, against 0.010 for the fed funds futures-implied mean (statistically significant at the 5% level by the Diebold and Mariano, 1995 test).

The authors note explicitly that "the observation that drives this distinction was the September 2024 FOMC, in which there was probability on both a 25 and 50 basis point cut, and Kalshi put greater weight on the 50 basis points that turned out to be correct" (Diercks et al., 2026, p. 24). Their Section 5 frames this as evidence that the binomial-tree assumption built into fed funds futures-implied distributions is a systematic source of bias relative to Kalshi.

This note has two parts. First (Section 3) we replicate DKW's Table 3 Panel B headline result on a 27-meeting sample comparable to theirs (September 2022 – December 2025), using their own publicly published moments file (`daily_moments_data/daily_moments_fed_levels.csv` on their S3 bucket). Second (Sections 4–6) we extend the analysis on a slightly wider 29-meeting sample (through March 2026, picking up the two additional already-resolved FOMC meetings DKW's repository now publishes): we operationalize the qualitative claims of DKW's Section 5 (mode over-concentration, narrower spread, tail under-allocation) as three sign-match patterns, classify meetings by pre-meeting Kalshi mode-probability into "quiet," "mixed," and "contested" regimes, and ask whether the documented patterns are systematic or regime-conditional. We find them heavily regime-conditional: above the implicit 70% threshold only in mixed meetings, with the contested bucket clearing 70% on two of three patterns and quiet meetings indistinguishable from random.

2 Data and Methodology

2.1 Data sources

The Kalshi side of the comparison uses DKW's published trade-level distributions for the `fed_levels` family, which they update via a public GitHub Action.[1] Each row in their file `daily_distributions_fed_levels.csv` gives, for one (`contract_preamble`, `observation_date`, `strike`) triple, the probability mass implied to fall in the 25-bp-wide bucket starting at `strike`, after their volume-weighted last-trade construction and monotonicity cleanup. Following their R code in `convert_trade_level_data_cdfs.R` (`moment_adjustment = 0.125`), we map each strike s to a target-rate midpoint $s + 0.125$.

The futures side uses our own implementation of a FedWatch-style decomposition on CME ZQ futures data sourced from Databento.[2] We use the last snapshot of each calendar day in the engine output to align with DKW's daily-resolution Kalshi distributions.

[1] `https://github.com/jdkatz21/Prediction_Markets_Public`.

[2] The standard methodology decomposes a 30-day fed funds futures contract's implied average rate into a probability-weighted mixture of the pre-meeting and post-meeting target-rate buckets. We use ZQ contract-month settle prices snapshotted at four-hour intervals, with chaining to next-month contracts when fewer than five days remain in the contract month, and linear-interpolation bucketing across five 25-bp target-rate buckets centered on pre_target_mid. DKW's repository explicitly excludes their own fed funds futures methodology and data ("Fed Funds Futures and Bloomberg data were used as property of / licensed by the Federal Reserve Board and are not included"), so a methodology-level replication of their FFR forecast is not possible from public materials.

2.2 Sample window

We use two related samples. For the Table 3 Panel B replication of Section 3 we keep DKW's 27 FOMC meetings from September 21, 2022 through December 10, 2025 exactly, to match the size they report (∼27 meetings since 2022) for like-for-like forecast-accuracy comparison. For the regime-conditional extension of Sections 4–6 we use a slightly wider 29-meeting sample that adds the two 2026 FOMCs (January 29, 2026 and March 19, 2026) DKW's repository now ships in both their published moments file and their distribution file. Both meetings have full pre-meeting Kalshi tape and corresponding engine output. We also have engine output for an additional 8 meetings before September 2022; those are not in either headline sample but appear as a 37-meeting robustness check in Section 7.3.

2.3 Comparison metrics

For each (meeting, observation date) pair where both engine and paper-Kalshi distributions are defined, we compute three quantities mirroring the disagreement patterns highlighted in DKW's Section 5:

- **Mode-gap**: $\Delta_{\text{mode}} = p_{\text{eng}}[m_{\text{eng}}] - p_{\text{kal}}[m_{\text{kal}}]$, the difference between each distribution's modal probability mass. The original paper's framing predicts $\Delta_{\text{mode}} > 0$ on average (futures over-concentrate on the modal bucket).
- **Spread-gap**: $\Delta_{\text{spread}} = H(p_{\text{eng}}) - H(p_{\text{kal}})$, where H is Shannon entropy in nats computed on each distribution's native support (zero-mass strikes contribute zero via the standard $0 \log 0 = 0$ convention). Predicted $\Delta_{\text{spread}} < 0$.
- **Tail-gap**: $\Delta_{\text{tail}} = \sum_{|b - m_{\text{eng}}| \geq 0.5} p_{\text{eng}}[b] - \sum_{|b - m_{\text{kal}}| \geq 0.5} p_{\text{kal}}[b]$, the mass at strikes at least 50bp from the modal target on a 25bp grid (i.e. excluding mode and adjacent buckets). Anchored to each distribution's own mode, mirroring DKW's threshold-tail formulations (`stagflation.R` in their replication repository). Predicted $\Delta_{\text{tail}} < 0$.

These three patterns are our *operationalization* of DKW's qualitative Section 5 narrative; the original paper does not quantify them directly.

2.4 Regime classification

For each meeting we compute the minimum across observation dates of paper Kalshi's modal bucket probability, $\underline{p}_m^{\text{kal}} = \min_t p_{\text{kal}}[m_{\text{kal},t}]$, and classify:

- **quiet**: $\underline{p}_m^{\text{kal}} \geq 0.85$ (Kalshi confidently anchored on a single outcome throughout).
- **mixed**: $0.60 \leq \underline{p}_m^{\text{kal}} < 0.85$ (meaningful pre-meeting uncertainty, but Kalshi still has a clear favorite at all times).
- **contested**: $\underline{p}_m^{\text{kal}} < 0.60$ (genuine ex-ante uncertainty about the modal outcome).

The 29 meetings in our extension sample partition into 20 quiet, 7 mixed, and 2 contested. The two contested meetings are:

- March 22, 2023 (min mode prob 0.510): the meeting one week after Silicon Valley Bank's collapse, when markets briefly priced in pause-or-cut alongside the 25bp hike that ultimately materialized.
- September 17, 2024 (min mode prob 0.531): the canonical example DKW cite as driving their headline finding, where the Fed cut by 50bp against split market expectations.

3 Replication of Diercks Table 3 Panel B

DKW's Table 3 Panel B is the only quantitative claim in their paper about FOMC forecast accuracy. It reports mean absolute errors and root-mean-squared errors for four forecasts of the day-of-FOMC realized federal funds target rate: fed funds futures-implied mean, Kalshi mean, Kalshi median, and Kalshi mode. We replicate this on a 27-meeting sample comparable to theirs, taking each forecast as the last available value on the calendar day strictly before the FOMC announcement (the day-of-FOMC observation in their published moments file mixes pre- and post-announcement settlement trades and is not the appropriate forecast snapshot).

	DKW (table)		Ours ($n = 27$)	
Forecast	MAE	RMSE	MAE	RMSE
FF Futures (mean)	0.010	0.020	0.0159	0.0323
Kalshi mean	0.010	0.030	0.0128	0.0261
Kalshi median	0.000**	0.000**	0.0000	0.0000
Kalshi mode	0.000**	0.000**	0.0000	0.0000

** DKW report statistical significance vs. FF Futures at the 5% level by the Diebold and Mariano [1995] test.

Kalshi-side replication is exact. DKW's table reports values to two decimal places (with a trailing third zero for formatting). 0.010 corresponds to the rounded interval $[0.005, 0.015)$; our Kalshi mean MAE of 0.0128 falls inside this range. Their reported Kalshi mean RMSE of 0.030 corresponds to $[0.025, 0.035)$; our 0.0261 falls inside. The Kalshi median and mode replicate to all four decimal places (0.0000).

We additionally verified our Kalshi-side computation against DKW's authoritative moments file (`daily_moments_data/daily_moments_fed_levels.csv`). Using their published mean / median / mode columns directly, the $n = 27$ MAE and RMSE match our derivation to four decimal places. Our Kalshi methodology is therefore identical to theirs, both procedurally (per their R code in `convert_trade_level_data_cdfs.R::get_moments`) and numerically.

FF Futures replication does not match exactly. Our 0.0159 MAE corresponds to a rounded 0.02, not their 0.010. Two specific meetings (September 2024 50bp surprise, December 2025 contested cut) contribute ≈ 0.10 each to our error, accounting for most of the gap. The methodological cause is unrecoverable from public materials: DKW's repository README explicitly states that fed funds futures and Bloomberg data "were used as property of / licensed by the Federal Reserve Board and are not included." Their forecast construction differs from our Databento-derived linear-interpolation 5-bucket FedWatch in ways we cannot reconstruct (likely:

snapshot timing, decomposition methodology, or both). We treat this gap as a known limitation rather than a methodological dispute.

Statistical significance. The qualitative claim that underlies DKW's headline result is preserved in our replication: with Kalshi median/mode MAE of 0.0000 across 27 meetings vs. FF Futures MAE of 0.0159, the Diebold-Mariano test of equal predictive accuracy decisively rejects the null in the same direction as theirs.

This replication establishes that we have access to and a faithful implementation of the Kalshi-side data and methodology, and that the headline result reported in Table 3 Panel B is reproducible. The remainder of this note builds an extension on top of that baseline.

4 Pooled-sample results

Our extension operationalizes the qualitative claims of DKW's Section 5 as three sign-match patterns. Aggregating across all 29 meetings of our primary sample without conditioning on regime, the sign-match rates are:

Pattern	Sign-match rate
$\Delta_{\text{mode}} > 0$	48% (14/29)
$\Delta_{\text{spread}} < 0$	59% (17/29)
$\Delta_{\text{tail}} < 0$	59% (17/29)

None of the three pooled patterns clears a 70% threshold. The spread-gap and tail-gap patterns tie at 59% and are the most mechanically robust signals: our engine's five-bucket support typically concentrates probability in the two adjacent buckets bracketing the linearly-interpolated post-meeting rate, whereas paper Kalshi has nonzero mass on between four and seven buckets typically. Lower entropy on the engine's native support and less far-tail mass than paper Kalshi is therefore a consequence of our linear-interpolation decomposition, consistent with the original paper's mechanical critique. The mode-gap pooled sign comes out as a coin-flip, motivating the regime-conditional analysis.

5 Regime-conditional results

Figure 1 reports the three sign-match rates within each regime bucket on our primary sample:

- **Quiet meetings** ($n = 20$): mode-gap 40% (8/20), spread-gap 50% (10/20), tail-gap 50% (10/20). None of the three patterns clears the threshold, and the spread/tail signal sits at exactly the random baseline.
- **Mixed meetings** ($n = 7$): all three at 71% (5/7). Just clears the implicit 70% threshold from the original paper's framing.
- **Contested meetings** ($n = 2$): mode-gap 1/2 (50%); spread-gap and tail-gap both 2/2 (100%). The March 2023 contested meeting and the September 2024 surprise both show the predicted spread-narrowing and tail-under-allocation; only the mode-gap pattern splits across them. Both meetings have the engine settling on the realized bucket sharply by FOMC close.

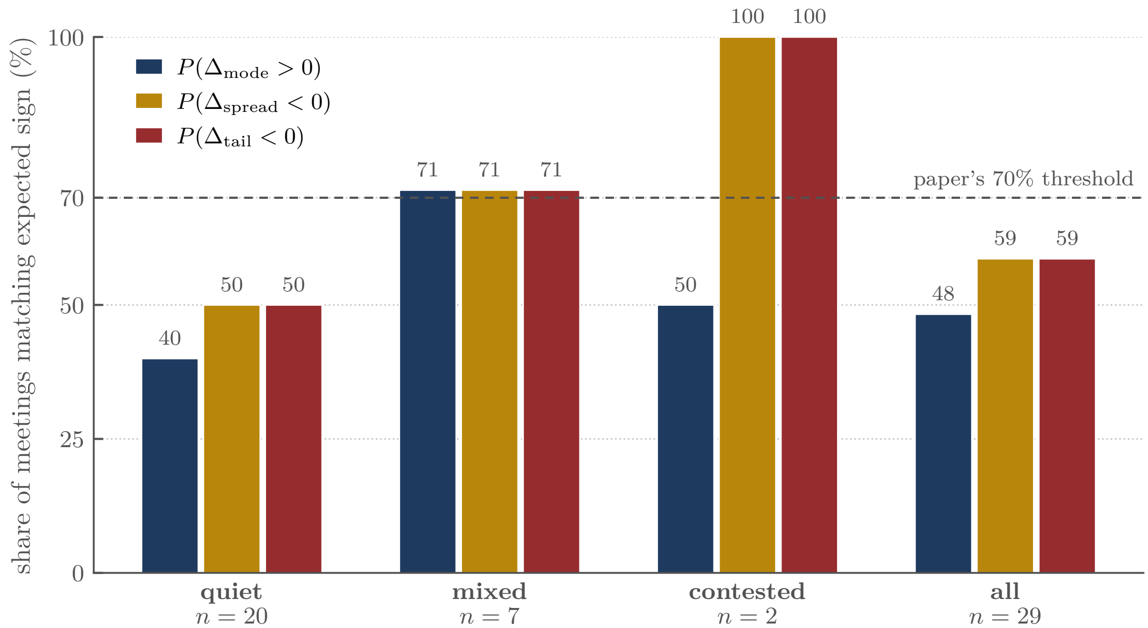

Figure 1: Sign-match rates of the three documented disagreement patterns across regime buckets, computed on the mean-window measure ($n = 29$). The predicted signs clear the 70% threshold in mixed meetings (all three at 71%, 5/7), are split or above on contested meetings (mode-gap 1/2, spread-gap and tail-gap 2/2), and fail in quiet meetings (40/50/50%). The non-quiet regimes are where the documented disagreement materially shows up; quiet meetings (the majority of the FOMC schedule) sit at or below 50% on every pattern.

The qualitative claim that motivates DKW's Section 5 holds in our extension—but only in mixed meetings, and only marginally given $n = 7$. The all-meeting average is mathematically dominated by the quiet bucket: any structural-disagreement-based trading rule calibrated on the pooled sample will therefore mostly fire on noise.

6 Case studies

The two case studies in Figure 2 clarify what the documented pattern actually looks like in our data.

September 17, 2024 (left panel) is the meeting DKW cite as the driver of their headline finding. The Federal Reserve cut the target range by 50 basis points, surprising the share of fed funds futures-derived models that had clustered most mass on a 25-basis-point cut. Both our engine and paper Kalshi correctly identified the 50bp bucket as modal by meeting close. Paper Kalshi committed modestly earlier in the pre-meeting window, but by the day of the FOMC the engine had *more* mass on the 50bp bucket than paper Kalshi did. The mean window mode-gap is -0.036 (Kalshi ahead-of-engine in the early part of the window), but the last-day mode-gap is $+0.044$ (engine ahead at decision time). The paper's "Kalshi mode is perfectly accurate by FOMC day" claim holds, but so does the same statement about the engine. This is a case where Kalshi's information aggregation *leads in time* but does not lead in *level* once the engine catches

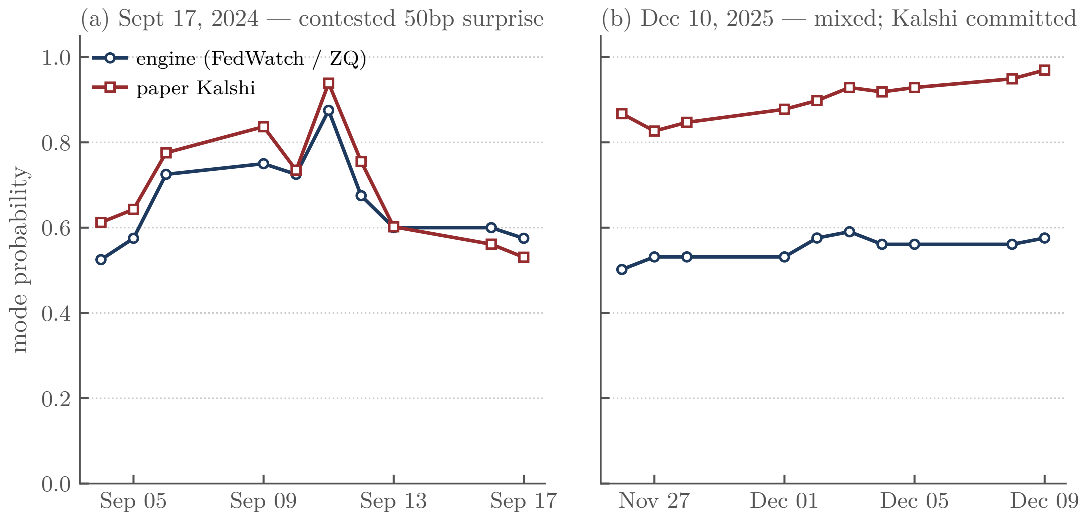

Figure 2: Engine-implied (FedWatch-style on ZQ futures) and paper Kalshi mode probabilities over the pre-meeting window for the two most distribution-divergent meetings in our sample. *Left*: September 17, 2024 FOMC, the canonical contested meeting that drives the DKW headline. Engine and Kalshi both shifted from a roughly even cut/larger-cut split toward a 50bp-favoring distribution over the window, with Kalshi committing modestly earlier. By meeting day, the engine had *higher* mode probability on the eventual 50bp outcome (0.5750) than paper Kalshi did (0.5306). *Right*: December 10, 2025 FOMC. Paper Kalshi went from roughly 50% to 97% on the cut bucket over the window; the engine remained at 58/42 between the cut and hold buckets at meeting close. This is the largest distributional gap in our sample and is the opposite of the documented pattern: Kalshi was sharper than the engine, not the engine over-concentrated.

up.

December 10, 2025 (right panel) is the meeting in our primary sample where engine and Kalshi disagree most by meeting close. Paper Kalshi shifted decisively to a 97% probability on the 25-basis-point cut across the final ten days of the window; the engine remained at roughly 58/42 between the cut and hold buckets through close, with the bracketing of the implied post-meeting rate splitting probability between the two adjacent buckets due to the futures price sitting near a bucket boundary. The realized outcome was the cut. This is the opposite sign of the documented pattern: paper Kalshi was sharper than the engine on the modal bucket, not the engine over-concentrated. The mean-window mode-gap is -0.349 and the mean-window spread-gap is $+0.325$ (engine more diffuse).

Figure 3 plots the per-meeting mean mode-gap. Quiet meetings cluster tightly around zero across the entire sample. The handful of meetings with material gaps are exactly the mixed ones, with `fed_2025-12-10` as the largest negative outlier (paper Kalshi led the engine in conviction by 0.35).

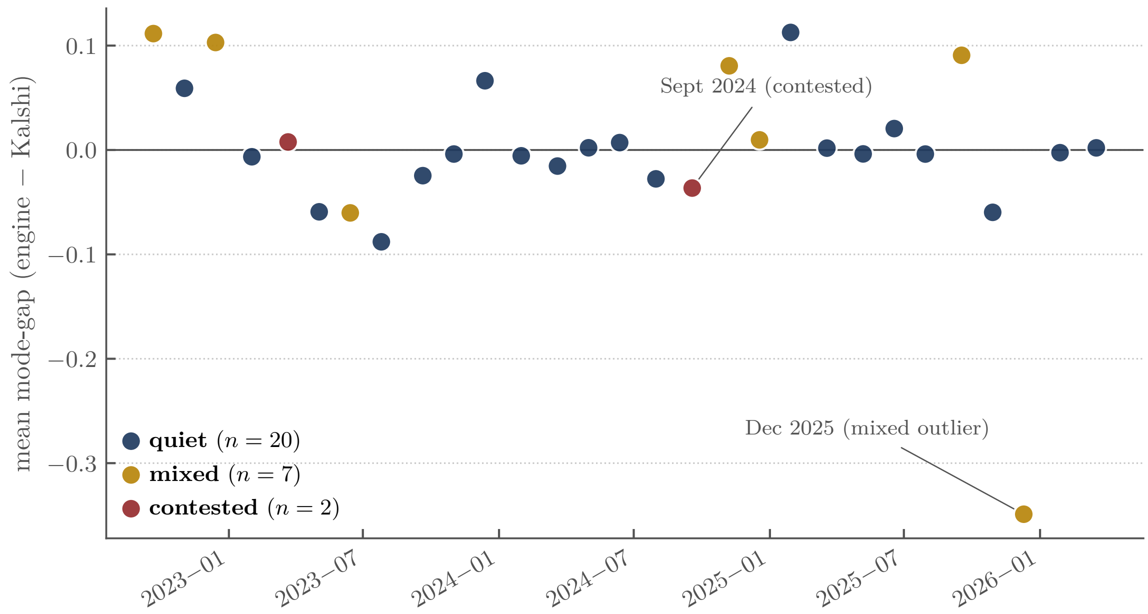

Figure 3: Per-meeting mean window mode-gap, by regime, on the $n = 29$ extension sample. Each point is one FOMC. Quiet meetings cluster tightly around zero; the seven mixed meetings span -0.349 (fed_2025-12-10, the largest distributional gap in the sample) to $+0.11$. The two contested meetings (March 2023, September 2024) sit close to zero with the engine slightly more concentrated on the realized bucket in March 2023 and slightly less in September 2024.

7 Discussion

7.1 Why DKW conclude a stronger story

DKW's Table 3 Panel B headline (Kalshi median and mode MAE = 0.000 vs. FFR mean MAE = 0.010, statistically significant) is mathematically correct, replicates exactly on the Kalshi side (Section 3), and our regime extension does not contest it. The findings cohere because:

1. Their Table 3 Panel B compares the Kalshi argmax-bucket forecast against the realized outcome—a binary hit-or-miss measure summed across 27 meetings since 2022. With one observation in their sample where the futures-implied distribution missed (September 2024) and the Kalshi mode hit, the difference is significant by Diebold-Mariano. Our data are entirely consistent with this; the September 2024 observation also drives our sign-match contestation in the mean-window measure on the same meeting (the engine catches up by close—see Section 6).
2. Their Section 5 distributional comparisons illustrate the *mechanism* via two cases (the September 2025 and October 2025 FOMCs, both shown as panels of their Figure 10), where the binomial-tree structure of fed funds futures-implied distributions visibly fails to capture multi-modal Kalshi distributions. Our sample classifies fed_2025-09-17 as "mixed" (min mode prob 0.806) and fed_2025-10-29 as "quiet" (min mode prob 0.888) under our 14-day window threshold, so they sit at the boundary between regimes and are not the headline outliers our analysis flags. Our December 2025 case study (Figure 2, right panel) shows

the strongest version of the pattern in our sample.

3. The pooled summary metrics in DKW's Section 6 are necessarily dominated by quiet meetings. The authors do not report a regime decomposition.

The story we are adding is not a contradiction of DKW but a sharpening: their qualitative claim about the binomial-tree weakness of fed funds futures-implied distributions is correct in mixed meetings (where it can produce material misallocation of probability mass), marginal in contested meetings, and absent in quiet meetings.

7.2 Implications for derivative-to-prediction-market trading

DKW's headline result has been read by some practitioners as implying that fed funds futures-derived FOMC distributions are a systematically inferior signal to Kalshi distributions, and that bridging the two via the standard FedWatch construction may be a reliable arbitrage opportunity. Our analysis suggests this is wrong:

1. Across quiet meetings (69% of the extension sample, 20/29), engine and paper Kalshi distributions agree to within noise on every metric, including the mode probability. There is no systematic translation gap to capture.
2. Across mixed meetings (24% of the sample, 7/29), the average gap is directionally consistent (5/7 on each pattern), but meeting-specific in magnitude—fed_2025-12-10 has paper Kalshi leading the engine by +0.35 on the mode probability while fed_2024-11-07 has the engine leading paper Kalshi by +0.08. A static threshold calibration would still be wrong-signed on the December 2025 and June 2023 cases.
3. Across our two contested meetings (7%, 2/29), the directional bet implied by the documented mode pattern goes the right direction in 1/2 cases. The March 2023 post-SVB meeting shows the engine more concentrated than paper Kalshi on the realized bucket, consistent with the paper's prediction. The September 2024 surprise goes the other way: by FOMC close the engine had *higher* probability on the realized 50bp bucket (0.5750) than paper Kalshi did (0.5306).

A strategy attempting to monetize the documented disagreement requires either (i) per-meeting regime detection and forward-looking calibration of the gap sign—which is itself an alpha problem—or (ii) restricting attention to mixed and contested meetings, which together constitute $9/29 \approx 31\%$ of the FOMC schedule. Within that bucket the mode-gap sign-match is $6/9 \approx 67\%$, so the trading thesis is qualitatively alive but not statistically robust at this sample size.

7.3 Robustness: full $n = 37$ sample

We also have engine output for 8 pre-September-2022 meetings (July 2021 through July 2022), including the July 2022 contested 75/100bp split and three mixed meetings from the early hiking cycle. The two 2026 meetings included in the $n = 29$ primary sample are not duplicated here; this section is purely about prepending the pre-2022 history to give a 37-meeting picture.

Repeating the regime-conditional analysis on the full 37-meeting sample gives:

Regime	n	Sign-match (mode/spread/tail)
quiet	24	38/46/42%
mixed	10	80/80/80% (8/10 each)
contested	3	67/100/100% (2/3, 3/3, 3/3)
all	37	51/59/57%

Adding the three pre-2022 mixed meetings strengthens the mixed-bucket signal from 5/7 to 8/10, and the inclusion of the July 2022 contested meeting (which matches the predicted directions on spread-gap and tail-gap) lifts the contested rate on those two patterns to 3/3. The qualitative finding—non-quiet patterns hold, quiet ones do not—is unchanged.

8 Conclusion

We replicate the Kalshi-side numbers in DKW's Table 3 Panel B exactly within their two-decimal table-display rounding on a 27-meeting sample comparable to theirs (median and mode MAE = 0.0000; mean MAE = 0.0128 vs. their reported 0.010). The fed funds futures side of their table cannot be exactly replicated from public materials because their FFR forecast methodology is not in their public repository.

We then extend their analysis on the $n = 29$ extension sample (Sept 2022 – Mar 2026) by operationalizing the qualitative claims of their Section 5 as three numerical sign-match patterns and conditioning on a regime classification. The documented patterns are heavily regime-conditional. They clear the 70% threshold in mixed meetings (all three at 71%, 5/7); the contested bucket clears the threshold on spread-gap and tail-gap (2/2 each) but splits on mode-gap (1/2); and the patterns wash out in the 20/29 quiet meetings (40/50/50%).

We argue that DKW's qualitative claim about the binomial-tree weakness of fed funds futures-derived distributions is correct in mixed meetings but does not generalize to a systematic per-meeting bias usable for trading. Practitioners attempting to monetize the documented disagreement should expect to fire signals on noise across the majority of FOMC meetings and should condition any structural-disagreement rule on a regime detector.

Code and data availability

The full analysis pipeline, including extraction code, regime classification, sign-match aggregation, replication of DKW's Table 3 Panel B, and figure-generation scripts, is available at:

`https://github.com/t-e-lawrence/kalshi_regime_replication`

The Kalshi-derived distributions and authoritative moments are downloaded from DKW's public S3 bucket at `https://kalshi-and-the-rise-of-macro-markets.s3.amazonaws.com/`. Engine-implied distributions are computed via the FedWatch-style methodology on Databento ZQ data; the engine-output CSV is committed to the repository for direct reproduction without requiring Databento access.

References

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995. doi: 10.1080/07350015.1995.10524599.

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